

COUNTING COLLINEAR TRIPLES, FOUR DISTINCT TRIANGLES FORCE AT MOST SEVEN POINTS, AND THE REGULAR HEPTAGON

ABSTRACT. Count a triangle as a triple of distinct points of the plane up to rigid motion, collinear triples included, as Palsson and Yu do. With triangles counted that way, and not under the rival convention noted below, we prove their Conjecture 4.1: a planar set spanning at most four distinct triangles has at most seven points, and a seven-point set spanning four distinct triangles is similar to the vertex set of a regular heptagon. The proof is a complete census of the 161 351 edge-colourings of K_7 that use at least three distance values and span at most four triangles, an elimination of all but one S_7 -orbit of them by exact polynomial certificates in integer and rational arithmetic, a Fourier computation identifying the surviving circulant as the regular heptagon, and a short hand argument for eight points and more. Under the rival convention that counts only non-collinear triples neither half of the conjecture is decided here, and the uniqueness half is false: the regular hexagon with its centre is a seven-point counterexample.

1. INTRODUCTION

For a finite set $S \subset \mathbb{R}^2$ let $T(S)$ be the number of equivalence classes of 3-element subsets of S under rigid motions of the plane. Palsson and Yu [4] fix this counting convention explicitly; we quote it verbatim:

Convention. “To avoid ambiguity regarding degenerate triangles, we can formalize this as counting the number of triples of distinct points in a point set, up to the equivalence relation defined by rigid plane isometries (hence, the three vertices of the triangle are allowed to be collinear, but must be distinct).”

So collinear triples are counted, and by side–side–side a class is exactly a multiset of three pairwise distances. Their Theorem 1.1 states that a planar set spanning at most three distinct triangles has at most six points, uniquely the regular hexagon, and their Section 4 then reads, verbatim:

“Firstly, observing that the regular heptagon has 4 distinct triangles, we conjecture that this is optimal.”

Conjecture 4.1. “The maximum number of points in the plane spanning four distinct triangles is 7, achieved only by the regular heptagon.”

2020 *Mathematics Subject Classification.* 52C10, 51M04, 52C35.

Key words and phrases. distinct triangles, few-distance sets, regular heptagon, Cayley–Menger determinant, exhaustive census, Schoenberg’s criterion.

Theorem 1.1. *Count triangles as in the Convention. Then*

- (a) *every set of seven points in the plane spans at least four distinct triangles;*
- (b) *a set of seven points in the plane spanning at most four distinct triangles is similar to the vertex set of a regular heptagon;*
- (c) *no set of eight or more points in the plane spans at most four distinct triangles.*

In particular $\max\{|S| : T(S) = 4\} = 7$, attained only by the regular heptagon: Conjecture 4.1 of [4] holds with triangles counted as in the Convention. It is not decided here under the rival convention described in the Scope paragraph below.

Part (a) reproves the seven-point case of [4, Theorem 1.1] and is used in the proof of (c). Parts (a) and (b) are Proposition 4.6 below; part (c) is Proposition 5.1.

Scope. Theorem 1.1 is proved under the Convention, the count [4] states. Under the rival convention that counts only non-collinear triples — used by [2, Definitions 3.2–3.3] and by [1] — this paper decides *neither* half of Conjecture 4.1, and the uniqueness half is then *false*: the regular hexagon together with its centre is a seven-point set with exactly four non-collinear classes which is not similar to a regular heptagon (Remark 2.2). Our census enumerates the cell “at most four classes *with* collinear triples counted”, and that configuration has five classes in our count, so it was never in the cell; for the same reason nothing here bears on the maximality half under the rival convention, which we leave open, nor on [2, Conjecture 2.1] (“any set of seven points in the plane determines at least four distinct triangles”), which is stated under the non-collinear count.

Lemmas 4.1–4.5 are finite computations, carried out in exact integer and rational arithmetic by the program accompanying this note; Sections 2, 3 and 5 are by hand, and Section 6 says what the program covers. The census splits by the number D of distinct distances, and 156 710 of its 161 351 leaves have $D = 3$; that lane is covered by Shinohara’s classification of planar three-distance sets [6], which is not used here. Enumerate colourings, then eliminate by realizability, is the method of Szöllősi and Östergård [7].

2. THE TWO CONFIGURATIONS

Let $\zeta = e^{2\pi i/7}$ and let $H = \{\zeta^0, \dots, \zeta^6\}$ be the regular heptagon. For $g \in \{1, 2, 3\}$ put

$$s_g = |1 - \zeta^g|^2 = 2 - 2\cos(2\pi g/7),$$

so s_1, s_2, s_3 are the three squared chord lengths. With $c = 2\cos(2\pi/7)$, which satisfies $c^3 + c^2 - 2c - 1 = 0$, one has $s_1 = 2 - c$, $s_2 = 4 - c^2$ and $s_3 = 1 + c + c^2$, and

$$(1) \quad s_1, s_2, s_3 \text{ are the three roots of } x^3 - 7x^2 + 14x - 7.$$

All three are positive, because for $x \leq 0$ every term of $x^3 - 7x^2 + 14x - 7$ is ≤ 0 and its constant term is $-7 < 0$; and they are distinct, that cubic being irreducible over $\mathbb{Q}$. Numerically $s_1 = 0.75302\dots$, $s_2 = 2.44504\dots$, $s_3 = 3.80193\dots$.

Lemma 2.1. *$T(H) = 4$ and H realises exactly three distinct distances. No triple of vertices of H is collinear, so $T(H) = 4$ under either convention.*

Proof. The chord $\zeta^i \zeta^j$ has squared length s_g with $g = \min(|i-j|, 7-|i-j|)$, and $g \mapsto s_g$ is injective by (1), so a triple's class is the multiset of the three cyclic gaps it cuts out. Those gaps are a partition of 7 into three positive parts read modulo the folding $g \mapsto \min(g, 7-g)$, and the four partitions $5+1+1$, $4+2+1$, $3+3+1$, $3+2+2$ give the four multisets

$$\{s_1, s_1, s_2\}, \quad \{s_1, s_2, s_3\}, \quad \{s_1, s_3, s_3\}, \quad \{s_2, s_2, s_3\},$$

and no others; all $\binom{7}{3} = 35$ triples are accounted for. A triple with squared side lengths a, b, c is collinear exactly when $2(ab+bc+ca) - (a^2+b^2+c^2) = 0$, that expression being sixteen times the squared area; it is a nonzero element of $\mathbb{Q}(c)$ for each of the four multisets above. $\square$

Remark 2.2. Let X consist of the six vertices of a unit regular hexagon together with its centre, say

$$(\pm 1, 0), \left(\pm \frac{1}{2}, \pm \frac{\sqrt{3}}{2}\right), (0, 0).$$

Its 21 squared distances take the three integer values 1, 3, 4, and its 35 triples fall into the five classes $\{1, 1, 1\}$, $\{1, 1, 3\}$, $\{1, 1, 4\}$, $\{1, 3, 4\}$, $\{3, 3, 3\}$. Exactly one of them, $\{1, 1, 4\}$, is collinear — a vertex, the centre and the opposite vertex — and it occurs for three triples, whereas H has none. So $T(X) = 5$ under the Convention, while X spans only four *non-collinear* classes on seven points and is not similar to H .

3. COLOURINGS, REALIZABILITY, AND TWO CERTIFICATES

Let $S = \{p_0, \dots, p_{n-1}\} \subset \mathbb{R}^2$ consist of n distinct points and let $d_{ij} = |p_i - p_j|^2$. Colour the edge $\{i, j\}$ of K_n by the *value* d_{ij} ; write D for the number of colours. By side-side-side, two triples are congruent if and only if their multisets of edge colours agree, so

(2)

$$T(S) = \#\{\text{distinct multisets of edge colours of the } \binom{n}{3} \text{ triangles of } K_n\}.$$

Number the edges $(0, 1), (0, 2), (1, 2), (0, 3), \dots, (n-2, n-1)$ and number the colours $1, 2, \dots$ in order of first appearance; this puts the colouring in a unique *restricted-growth word* $w(S) \in \{1, \dots, D\}^{\binom{n}{2}}$, and it quotients the relabelling of colours exactly. Scaling S scales all d_{ij} together, so we may and do normalise the squared distance of colour 1 to 1 and write $y_2, \dots, y_D > 0$ for the others; these are pairwise distinct and distinct from 1, since distinct colours are distinct distance values.

We use one classical criterion and two elementary consequences of it.

Lemma 3.1 (Schoenberg [5]). *Let $M = (m_{ij})$ be a real symmetric $n \times n$ matrix with zero diagonal and fix b . There are points $p_0, \dots, p_{n-1} \in \mathbb{R}^k$ with $|p_i - p_j|^2 = m_{ij}$ for all i, j if and only if the $(n-1) \times (n-1)$ matrix*

$$G^{(b)} = \left(\frac{1}{2} (m_{bi} + m_{bj} - m_{ij}) \right)_{i,j \neq b}$$

is positive semidefinite of rank at most k . ($G^{(b)}$ is the Gram matrix of the vectors $p_i - p_b$.)

We shall use the criterion in the following equivalent form. Let $H_0 = \{x \in \mathbb{R}^n : \sum_i x_i = 0\}$. If $m_{ij} = |p_i - p_j|^2$ and $\sum_i x_i = 0$ then

$$x^\top M x = \sum_{i,j} x_i x_j (|p_i|^2 - 2\langle p_i, p_j \rangle + |p_j|^2) = -2 \left| \sum_i x_i p_i \right|^2,$$

and $\sum_i x_i p_i = \sum_{i \neq b} x_i (p_i - p_b)$, so under $x \mapsto (x_i)_{i \neq b}$ the form $-M$ restricted to H_0 is twice the quadratic form of $G^{(b)}$. In particular $-M|_{H_0}$ is positive semidefinite of rank exactly $\text{rank } G^{(b)}$, so realizability in $\mathbb{R}^k$ says: $-M$ is positive semidefinite on H_0 with at most k nonzero eigenvalues there.

Corollary 3.2. *If $S \subset \mathbb{R}^2$ then, for every b , every 3×3 minor of $G^{(b)}$ vanishes; equivalently, every 4-point Cayley–Menger determinant*

$$\text{CM}(i, j, k, l) = \det \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & d_{ij} & d_{ik} & d_{il} \\ 1 & d_{ij} & 0 & d_{jk} & d_{jl} \\ 1 & d_{ik} & d_{jk} & 0 & d_{kl} \\ 1 & d_{il} & d_{jl} & d_{kl} & 0 \end{pmatrix}$$

vanishes.

Fix a restricted-growth word w on K_n . Each $\text{CM}(i, j, k, l)$ and each 3×3 minor of each $G^{(b)}$ is then a polynomial with integer coefficients in $y_2, \dots, y_D$, computable from w alone; call these the *test polynomials* of w . By Corollary 3.2 all of them vanish at any planar realization of w , whose coordinates y lie in the open positive orthant. Hence:

Lemma 3.3 (one-signed certificate). *If some test polynomial of w is nonzero and all of its coefficients have the same sign, then w has no planar realization.*

Proof. All exponents are non-negative, so such a polynomial is strictly positive, or strictly negative, at every point of the positive orthant, and in particular does not vanish there. $\square$

Lemma 3.4 (univariate gcd certificate). *Let U be the set of test polynomials of w that involve the single variable y_i , and let $g = \gcd U \in \mathbb{Q}[y_i]$, taken monic. Then $g(y_i) = 0$ at any planar realization. Consequently, writing $g = y_i^k h$ with $h(0) \neq 0$:*

- (i) if h is constant and $k = 0$, then w has no planar realization;
- (ii) if h is constant and $k \geq 1$, then $y_i = 0$, which is impossible for distinct points, so w has no planar realization;
- (iii) if $\deg h = 1$ and its unique root r is positive, then $y_i = r$, and substituting $y_i = r$ into every test polynomial of w leaves a system in the remaining variables to which (i)–(iii) may be applied again.

Proof. Every element of U vanishes at a realization, hence so does any polynomial combination of them, hence so does g . In (iii), $y_i > 0$ forces $y_i = r$ since r is the only root of h and $y_i^k \neq 0$. $\square$

4. SEVEN POINTS

Throughout this section $n = 7$, so K_7 has 21 edges and $\binom{7}{3} = 35$ triples, and the *cell* is the set of restricted-growth words on K_7 with at most four distinct triple colour-multisets: by (2) the word of any seven-point set with $T(S) \leq 4$ lies in the cell.

Lemma 4.1. *Exactly 161 351 words of the cell use $D \geq 3$ colours. Every one of them has $D \leq 4$; 156 710 have $D = 3$ and 4641 have $D = 4$. Split further by the number of triple multisets realised, the counts are 3948 with $(D, T) = (3, 3)$, 152 762 with $(3, 4)$ and 4641 with $(4, 4)$.*

Proof. A depth-first traversal that assigns colours to the 21 edges in order, maintaining the set of triple multisets already completed and pruning as soon as a fifth one appears. The traversal is complete and uncapped: it explores 2 915 582 nodes and terminates. That $3948 + 152\,762 + 4641 = 161\,351$ is the arithmetic of the exhaustion. $\square$

Lemma 4.2. *Of those 161 351 words, 81 306 have a 4-subset whose Cayley–Menger determinant is a nonzero constant, and 70 286 further ones have a 4-subset whose Cayley–Menger determinant is non-constant with all coefficients of one sign. By Lemma 3.3 none of these 151 592 words is realizable. The remaining 9 759 words form exactly 30 orbits under the action of S_7 on the vertices.*

Proof. A finite computation, 35 determinants per word; $81\,306 + 70\,286 + 9\,759 = 161\,351$. The test depends on the word only through its 4-subsets, so the surviving set is S_7 -stable, and the orbit count is a union-find over two generators of S_7 . The orbit sizes, in the notation size^{multiplicity}, are $70, 105^5, 120, 140, 210^5, 252^2, 315^8, 420, 630^5, 1260$, and they sum to 9 759. $\square$

Lemma 4.3. *Of those 30 orbits, 27 carry a 3×3 Gram minor that is nonzero with all coefficients of one sign, and two more are killed by Lemma 3.4. Exactly one orbit remains, the orbit of size 120 of the word*

$$(3) \quad W = 112213231332312332211,$$

listed against the edge order

01, 02, 12, 03, 13, 23, 04, 14, 24, 34, 05,
15, 25, 35, 45, 06, 16, 26, 36, 46, 56.

Proof. For 27 orbits, Lemma 3.3. The two others have $D = 4$, and for each of them we record the monic gcd's of Lemma 3.4 in the order the substitutions are made. For 1 1 2 1 3 4 1 4 3 2 2 1 1 1 3 1 1 1 1 4 (orbit size 210): the 8 test polynomials in y_2 alone have gcd $y_2^3 - 2y_2^2$, so $y_2 = 2$; then the 32 in y_3 alone have gcd $y_3^2 - 2y_3$, so $y_3 = 2$; then the 2192 in y_4 alone have gcd y_4 , forcing $y_4 = 0$, which case (ii) of Lemma 3.4 excludes. For 1 1 2 2 1 1 3 4 4 3 3 4 4 3 2 4 3 3 4 1 1 (orbit size 105): gcd $y_2^3 - 2y_2^2$ gives $y_2 = 2$; gcd $y_3 - 1$ gives $y_3 = 1$; and then the 2288 test polynomials in y_4 alone have gcd 1, so by case (i) they have no common root at all. (The value $y_3 = 1$ already contradicts $y_3 \neq 1$, so that orbit dies twice over.) Finally $27 + 2 + 1 = 30$. $\square$

Lemma 4.4. *A seven-point planar set whose word is W is similar to H .*

Proof. Relabelling the vertices by $\sigma = (0, 1, 6, 2, 5, 3, 4)$, meaning $0 \mapsto 0$, $1 \mapsto 1$, $2 \mapsto 6$, $3 \mapsto 2$, $4 \mapsto 5$, $5 \mapsto 3$, $6 \mapsto 4$, carries W to the gap colouring of $\mathbb{Z}_7$, in which the colour of $\{i, j\}$ is $\min(|i - j|, 7 - |i - j|)$; there are $42 = |\text{AGL}(1, 7)|$ such relabellings, consistent with the orbit size $5040/42 = 120$. Each colour class of W is a single 7-cycle; colour 1, for instance, is 0 1 3 5 6 4 2.

So the squared-distance matrix is the circulant M on $\mathbb{Z}_7$ with $M_{ij} = y_g$, $g = \min(|i - j|, 7 - |i - j|)$, for positive y_1, y_2, y_3 . Its eigenvectors are the Fourier modes $v_m = (\omega^{jm})_j$, $\omega = e^{2\pi i/7}$, with

$$\mu_m = \sum_{g=1}^3 y_g c_{gm}, \quad c_k := 2 \cos(2\pi k/7) = c_{7-k},$$

so that $v_0 = (1, \dots, 1)$ has eigenvalue $\mu_0 = 2(y_1 + y_2 + y_3) > 0$. The modes $v_1, \dots, v_6$ span H_0 , and $\mu_m = \mu_{7-m}$, so Lemma 3.1, in the equivalent form derived just after it, says that realizability in $\mathbb{R}^2$ amounts to $\mu_m \leq 0$ for $m = 1, \dots, 6$ with at most two of them nonzero. Nonzero eigenvalues come in equal pairs, so their number is even, hence 0 or 2; it is not 0, since $M \neq 0$. Therefore exactly one of μ_1, μ_2, μ_3 is negative and the other two vanish.

Say $\mu_2 = \mu_3 = 0$. These are two linear equations in (y_1, y_2, y_3) over $\mathbb{Q}(c)$, and the 2×2 minor $c_2 c_1 - c_3 c_3$ of their coefficient matrix is a nonzero element of that field, so their solution space is a single line. The vector (s_1, s_2, s_3) lies on it and gives $\mu_1 = -7$, both by direct computation in $\mathbb{Q}(c)$; hence it spans it, and $(y_1, y_2, y_3) = \lambda(s_1, s_2, s_3)$ with $\lambda > 0$. By Lemma 3.1 the point set is then determined up to isometry, and it is $\sqrt{\lambda} H$.

The two remaining cases are the same set relabelled. The multiplier $j \mapsto 2j$ of $\mathbb{Z}_7$ sends gap g to gap $2g$, hence permutes $(1, 2, 3)$ as $(2, 3, 1)$, and $j \mapsto 3j$ as $(3, 1, 2)$; correspondingly $(y_1, y_2, y_3) \propto (s_2, s_3, s_1)$ gives $(\mu_1, \mu_2, \mu_3) =$

$(0, -7, 0)$ and (s_3, s_1, s_2) gives $(0, 0, -7)$, again by computation in $\mathbb{Q}(c)$. Both are the regular heptagon with its three distance classes renamed. $\square$

Lemma 4.5. *No seven-point planar set has $D \leq 2$.*

Proof. $D = 1$ is impossible already for four points: the Cayley–Menger determinant of a monochromatic 4-subset is a nonzero constant. For $D = 2$ it suffices, by heredity, to rule out six points. There are $2^{14} - 1 = 16\,383$ restricted-growth words on the 15 edges of K_6 using exactly two colours, and all of them lie in the cell, two colours yielding at most four triple multisets automatically. The Cayley–Menger test of Lemma 3.3 leaves 11 242 of them, forming 42 orbits under S_6 ; of these, 33 carry a one-signed 3×3 Gram minor and the remaining 9 are killed by Lemma 3.4. So no planar six-point two-distance set exists, which is the upper half of Erdős and Fishburn’s $g(2) = 5$ [3]. $\square$

Proposition 4.6. *Every seven-point planar set spans at least four distinct triangles, and one spanning at most four is similar to H .*

Proof. Let $|S| = 7$ with $T(S) \leq 4$. By Lemma 4.5, S has $D \geq 3$ distinct distances, so by (2) its word is one of the 161 351 words of Lemma 4.1; by Lemmas 4.2 and 4.3 that word lies in the orbit of W ; by Lemma 4.4 S is similar to H . That gives (b), and $T(H) = 4$ by Lemma 2.1 gives (a): a seven-point set with $T(S) \leq 3$ would be similar to H and so have $T(S) = 4$, a contradiction. $\square$

5. EIGHT POINTS AND MORE

Proposition 5.1. *No planar set of eight or more points spans at most four distinct triangles.*

Proof. Suppose $|S| = 8$ and $T(S) \leq 4$. For $p \in S$ the classes of $S \setminus \{p\}$ are among those of S , so $T(S \setminus \{p\}) \leq 4$; and $T(S \setminus \{p\}) \geq 4$ by Proposition 4.6(a). Hence $T(S \setminus \{p\}) = 4$ and, by Proposition 4.6(b), $S \setminus \{p\}$ is the vertex set of a regular heptagon, for every $p \in S$.

Pick $p \neq q$ in S and put $R = S \setminus \{p, q\}$, so $|R| = 6$ and $R \cup \{q\}$, $R \cup \{p\}$ are both regular heptagons. Six vertices of a regular heptagon determine the seventh. Indeed they are concyclic and no three are collinear, so the circle through any three of them is the circumcircle and is determined by R ; the six points cut that circle into six arcs, of which exactly one has length $2 \cdot (2\pi/7)$ and the others $2\pi/7$; and the missing vertex is the midpoint of the long arc. Applying this to R inside $R \cup \{p\}$ and inside $R \cup \{q\}$ gives $p = q$, a contradiction.

If $|S| \geq 9$ and $T(S) \leq 4$, any eight-point subset S' has $T(S') \leq 4$ by the same heredity, contradicting the previous paragraph. $\square$

Theorem 1.1 follows: (a) and (b) are Proposition 4.6, (c) is Proposition 5.1, and $T(H) = 4$ with $|H| = 7$ shows the maximum is attained.

6. VERIFICATION

The objects the hand arguments use are printed above: the heptagon and its four classes in Lemma 2.1, the hexagon-plus-centre in Remark 2.2, the surviving word W in (3) with its edge order and the relabelling σ that turns it into the gap colouring of $\mathbb{Z}_7$, the two eliminated $D = 4$ words with their gcd chains in Lemma 4.3, and the squared chords in (1). The census itself, the 161 351 words and the Cayley–Menger counts of Lemma 4.2, the 27 one-signed Gram minors of Lemma 4.3 and both lanes of Lemma 4.5 are not printed here; they are supplied through the program accompanying this note.

That program recomputes, in exact integer and **Fraction** arithmetic and using the Python standard library only, every numerical claim above: the squared chords and their minimal polynomial and the four heptagon classes with their non-degeneracy, over $\mathbb{Q}[c]/(c^3 + c^2 - 2c - 1)$; the five classes and one collinear class of the hexagon-plus-centre, over $\mathbb{Q}$; the census of Lemma 4.1 including its node count and its cross-tabulation; the three counts, the 30 orbits and their size profile in Lemma 4.2; the $27+2+1$ split of Lemma 4.3 with the gcd chains printed; the relabelling σ , the eigenvalue vectors $(-7, 0, 0)$, $(0, -7, 0)$, $(0, 0, -7)$ and the rank-two minor of Lemma 4.4; and both lanes of Lemma 4.5. It prints one line per check and exits 0 only if all of them pass; all 48 passed.

Nothing else is machine-checked. Schoenberg’s criterion (Lemma 3.1) is quoted, not proved; Proposition 5.1 contains no computation; and beyond the single configuration of Remark 2.2 nothing about the non-collinear convention is checked. The two certificates are sufficient conditions for non-realizability, and no claim is made that the 30 orbits could not be eliminated another way.

REFERENCES

- [1] H. N. Brenner, J. S. Depret-Guillaume, E. A. Palsson, and S. Senger, *Uniqueness of optimal point sets determining two distinct triangles*, *Integers* **21** (2021), Paper No. A43; arXiv:1910.00629. Its triangle count is the non-collinear one.
- [2] A. Epstein, A. Lott, S. J. Miller, and E. A. Palsson, *Optimal point sets determining few distinct triangles*, *Integers* **18** (2018), Paper No. A16. Definitions 3.2–3.3 use the non-collinear count; Conjecture 2.1 is stated there.
- [3] P. Erdős and P. Fishburn, *Maximum planar sets that determine k distances*, *Discrete Math.* **160** (1996), 115–125.
- [4] E. A. Palsson and E. Yu, *On optimal point sets determining distinct triangles*, *Electron. J. Combin.* **31** (2024), no. 2, Paper No. P2.24. doi:10.37236/12360; arXiv:2308.13107. Quotations above are verbatim from the source of arXiv:2308.13107v2.
- [5] I. J. Schoenberg, *Remarks to Maurice Fréchet’s article “Sur la définition axiomatique d’une classe d’espaces distanciés vectoriellement applicable sur l’espace de Hilbert”*, *Ann. of Math.* (2) **36** (1935), 724–732.
- [6] M. Shinohara, *Classification of three-distance sets in two dimensional Euclidean space*, *European J. Combin.* **25** (2004), 1039–1058. doi:10.1016/j.ejc.2003.12.009.

- [7] F. Szöllősi and P. R. J. Östergård, *Constructions of maximum few-distance sets in Euclidean spaces*, Electron. J. Combin. **27** (2020), no. 1, Paper No. P1.23. doi:10.37236/8565.